Surface and subgroup checks over GF(2^83)

2026-10-12 — run-20261012T0331Z

Result and scope

Both pinned Koblitz variants passed the native quartic-model and subgroup checks. Each embeds into its own product and into the matching factor of $E_0 \times E_1$, with the group law on the whole product. The opposite pure product fails the necessary transfer conditions. These are faithful representations of the original ECDLP; a faster solver was not supplied or measured. The default targets were public known-answer $Q = 37P$.

The field is $\mathbf{F}_{2^{83}}$, with polynomial basis $z^{83} + z^{45} + z^2 + z + 1$. The two sources are $E_a : y^2 + xy = x^3 + ax^2 + 1$, $a \in \{0, 1\}$. This is an arithmetic correctness run on one ordinary Linux x86-64 host, not an IC/rho benchmark.

Variant	Prime subgroup order r	Cofactor h
$a = 0$	241785163923079621668568 9	4
$a = 1$	8569786107849059	1128547018

Field degree 83 does not specify subgroup size. The $a = 0$ order has bit length 82; the $a = 1$ order has bit length 53. The existing Sage proof receipt is reused for both primes; the run additionally checks $a = 1$ with deterministic u64 Miller–Rabin. No new Sage primality run or second-host replay is claimed.

What the checker actually ran

The modulus passed Rabin's irreducibility criterion. The Frobenius trace recurrence reproduced the recorded orders and rh. Both nonidentity generators passed $rP = O$; both targets passed $rQ = O$ and the witness check. Each source and target had Frobenius period 83. Per variant the checker tested 166 orbit-point entries, each in its normalized and a second projective scale, plus eight transported-addition pairs. A point on a smooth containing quadric, outside the curve image, was rejected.

The same public targets were checked again with no scalar supplied: subgroup membership and the transported representation passed, and scalar-verification fields remained null. This tests the checker interface; the target values remain known-answer fixtures, not recovered unknown logarithms.

Four CLI controls exited with status 2: unsupported degree 82, the two-torsion point $(0, 1)$ in each odd-order subgroup, and an incorrect scalar witness. The native package's 22 tests passed. Its controls include independent three-word carryless multiplication and long reduction, exhaustive curve counts over degrees 1–5, and direct projective quadric counts over GF(2) and GF(4). The certificate package's five tests passed, including rejection of false integer lifts and the omitted characteristic-two correction.

Why the characteristic-two pencil works

Use the embedding and its inverse

$$(x, y) \longmapsto [1 : x : y : x^2], \qquad O \longmapsto [0 : 0 : 0 : 1],$$

$$Q_1 = WZ + X^2, \qquad Q_2 = Y^2 + XY + XZ + aWZ + W^2.$$

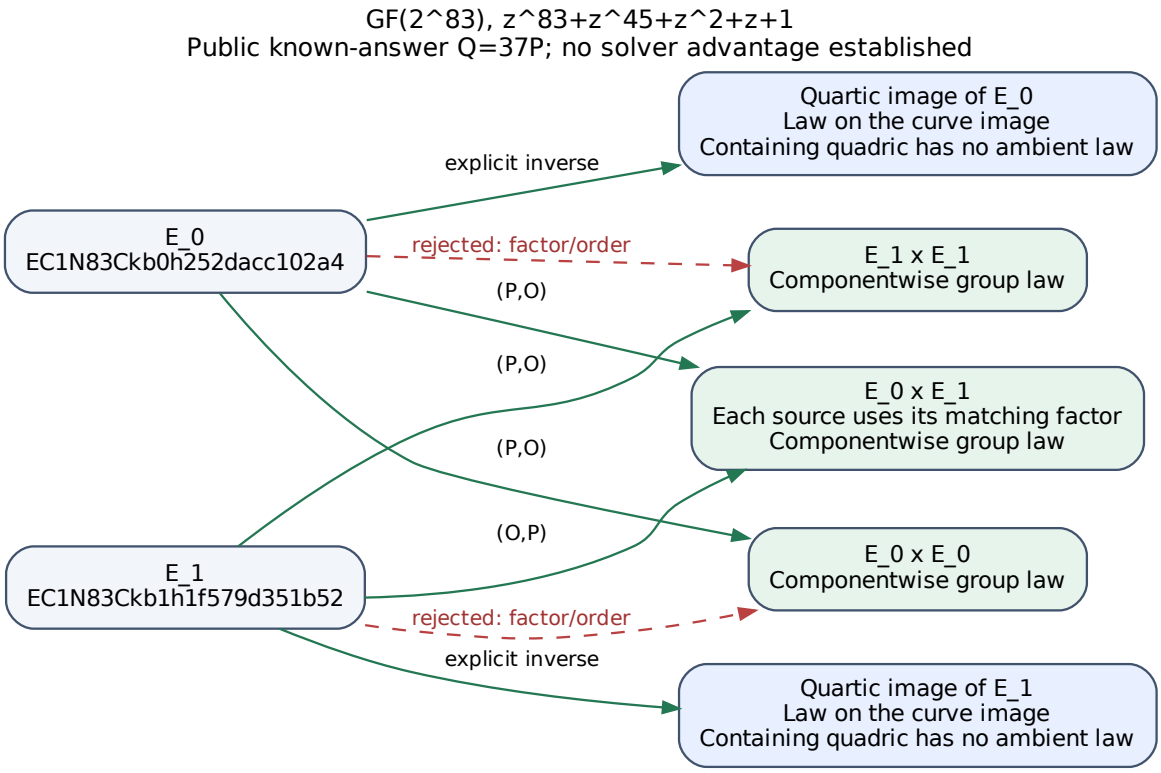


Figure 1: Verified inclusions and rejected opposite products. The source labels are exact representation IDs; see the records below.

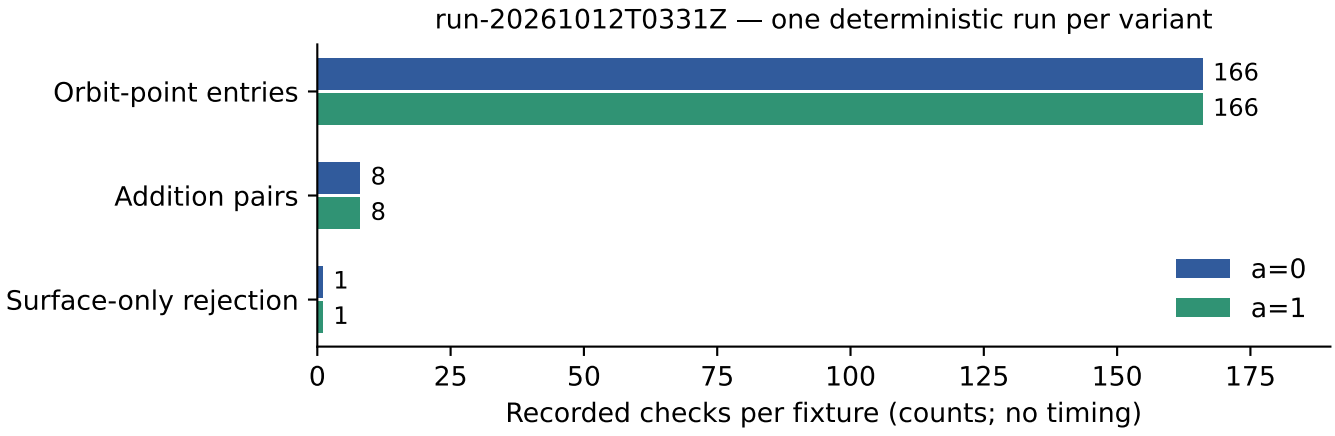


Figure 2: Recorded verification counts, one correctness run per variant. Entries are test populations, not distinct group states or performance measurements.

Their intersection is the curve image. On $W \neq 0$, its inverse is $(X/W, Y/W)$; on $W = 0$, the equations force $X = Y = 0$, leaving only the origin. The transported law is encode(decode(U) + decode(V)). Thus the inverse preserves the source subgroup and its ECDLP relation. A containing quadric's other points are outside that law's domain. The control $[1 : 0 : 0 : 1]$ lies on the smooth member $\lambda = a + 1$ but has $Q_1 = 1$, so decoding rejects it.

For $S_\lambda = Q_2 + \lambda Q_1$, put $c = \lambda + a$. Its polar matrix has determinant c^2. It is nonsingular for $c \neq 0$. At $c = 0$ the point $[1 : 0 : 1 : 1]$ is singular. The infinite member Q_1 is also singular. Hence exactly $q - 1$ finite members are smooth, where $q = 2^{83}$.

With $c \neq 0$, an invertible change of coordinates gives a hyperbolic plane and the binary form

$$t^2 + st + \delta_\lambda s^2, \qquad \delta_\lambda = \lambda + c^{-2}.$$

The member is split precisely when this binary form is isotropic, equivalently when $u^2 + u = \delta_\lambda$ has a solution. The Artin–Schreier map has kernel $\{0, 1\}$ and image equal to the absolute-trace-zero hyperplane. For odd degree 83 the trace map is nonzero since $\mathrm{Tr}(1) = 1$.

For each nonzero curve coordinate $x = c$, writing $y = xv$ gives $v^2 + v = x + a + x^{-2} = \delta_{a+x}$. There are two solutions exactly for a split member. At $x = 0$ there is one point $(0, 1)$, plus O. Therefore

$$\#E_a = 2 + 2\,\#\{\text{smooth split finite members}\}.$$

The curve is defined over GF(2). With $\tau_0 = 2$ and $\tau_1 = -1$ for $a = 0$, $\tau_1 = 1$ for $a = 1$, its extension traces satisfy $\tau_n = \tau_1\tau_{n-1} - 2\tau_{n-2}$ and $\#E_a = q + 1 - \tau_{83}$. These exact counts use algebra and the recurrence; the run does not enumerate 2^{83} parameters.

Variant	$\#E_a$	τ_{83}
$a = 0$	967140655692318486674 2756	-6151469093347
$a = 1$	967140655691088192855 6062	6151469093347

For both variants there are **967140655691703339764 9407** smooth finite members.

Variant	Smooth split	Smooth nonsplit
$a = 0$	4835703278461592433371377	4835703278455440964278030
$a = 1$	4835703278455440964278030	4835703278461592433371377

Product surfaces and the actual obstruction

Inclusion into $E_a \times E_a$ is $P \mapsto (P, O)$; projection recovers P. Inclusion into $E_0 \times E_1$ uses the matching component. Both products have componentwise elliptic addition on every point, and both inclusions have trivial kernel. Their Frobenius polynomials contain the source factor by construction. They leave the ECDLP on its original elliptic factor.

The source factors are $\chi_i(T) = T^2 - \tau_i T + q$ over the integers. Their difference is $(\tau_1 - \tau_0)T$, nonzero here, while $\chi_i(0) = q \neq 0$. Thus they are coprime and χ_a does not divide χ_{1-a}^2. Also r_0 does not divide $\#E_1$ and r_1 does not divide $\#E_0$; the recorded remainders are positive. Since these r are prime, neither can divide the opposite order squared. Hence the opposite product cannot receive an injective transfer of that prime subgroup over this field.

Frobenius polynomials retain integer coefficients and trace signs; their coefficients are not reduced modulo 2. The separate general polynomial CLI still has u64/i128 limits. This n=83 check handles explicit products, not arbitrary supplied large-q polynomials or new Jacobian constructions.

Identity records, representation IDs, and replay

Six `identity.certificate/v1` records each passed exact integer-coefficient expansion, 32 statement-derived evaluations and 32 additional evaluations. The four geometric lifts reduce modulo 2. The normal-form integer lift carries an explicit

$2bcus$ correction, which vanishes after reduction. The two Frobenius identities remain over the integers. The v1 evaluator remains over $2^{61} - 1$; these records do not claim random evaluation in GF(2^83), a formal proof, or a second host.

Identity	Certificate
Quartic embedding	`IDC1h6d8ec6e3d63f147b`
Alternating polar determinant	`IDC1hb0891fcca2c98751`
Cleared-denominator normal form	`IDC1hb857d55c10efa5ac`
Point-count substitution	`IDC1h0869c94c259c1d1a`
Frobenius difference	`IDC1h4d9709f79f94a6db`
Frobenius product	`IDC1h240f8c156bb16781`

Exact record IDs are `EC1N83Ckb0h252dacc102a4` and `EC1N83Ckb1h1f579d351b52`. Their full UID prefix is `urn:ec-record:1:sha256:` followed by these digests:

`252dacc102a4f15413d4a269e2156852ad18dd6f8b25cf207146868c4592e2cc`

`1f579d351b52b0f3aba9259aa9b77738a958c3dbb995a6965bb50822ad199bb3`

The archived a=1 benchmark uses another polynomial basis and point encoding. Its identifiers and targets are not substituted here. These new exact metadata records preserve their own hash preimages; no archived alias is reassigned.

From the repository root, reproduce the run with:

```
cargo run --release --offline --locked \
  --manifest-path tools/ecdlp-surfaces/Cargo.toml \
  --bin binary_surface_check -- --degree 83 --a both --output /tmp/n83.json
cargo run --release --offline --locked \
  --manifest-path docs/ecdlp-surfaces/certificates/Cargo.toml \
  --bin binary_identities -- /tmp/n83.json /tmp/n83-identities
tools/ecdlp-surfaces/check.sh
```

Raw evidence: both cases, no-witness a=0, no-witness a=1, identity replay, a=0 curve record, a=1 curve record, source and binary hashes, and the run manifest.

The full `crypto` library gate was attempted and remains blocked by its preexisting syntax error in the Koblitz factor-base source, line 1854 (diagnostic). Its source is unchanged. Conductor is unavailable. An initial launch used a root-relative manifest from the crate directory and failed with exit 101; the corrected commands above passed. That failure and the preliminary smoke output are retained separately from the final recorded evidence.

No IC candidate, isogeny route, performance row, or subgroup-size promotion is made. Canonical graph and registry checks are recorded in `GRAPH_REVIEW.md`. Arbitrary surface homomorphisms, target-solver cost, and any ECDLP advantage remain unresolved. No task is scheduled after this report.

Sources

- SEC 1 v2, §§2.1.2 and 2.2.2: polynomial-basis binary arithmetic and the elliptic addition formulas used here.
- Pinned n=83 fixtures: exact source generators, orders and cofactors.
- Existing prime-proof receipt and checked-Sage source: reused proof evidence; local copies and hashes accompany this study.
- Sutherland, MIT 18.783, Lecture 13: Frobenius characteristic-polynomial framework. The extension trace recurrence and binary classification are derived explicitly above.
- Native checker, independent native controls, and integer-lift checker.